Alexander Yao

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Education

University of Maryland College Park, MD
Bachelor of Science in Computer Science, Minor in Applied Mathematics *Aug. 2023 – (Expected) Dec. 2026*

- **Relevant Coursework**: Compilers, Parallel Computing, Advanced Data Structures, Algorithms, Computer and Network Security, Programming Languages, Computer Systems, Web Development, Probability and Statistics

Experience

Computer Network Research Assistant May 2025 – Feb. 2026
George Mason University *Fairfax, VA*

- Engineered a high-throughput media delivery system in Python (AsyncIO + QUIC) achieving sub-150ms end-to-end latency for 20 Mbps real-time video streams, reaching near-broadcast-grade SLA thresholds.
- Designed a custom congestion control and packet scheduling algorithm that cut packet loss by 50% and improved stream stability by 30% under high-loss conditions, validated across 20+ simulation runs.
- Built a bidirectional Python/Android client-server architecture over QUIC, enabling uninterrupted media delivery across variable network conditions.
- Deployed an observability pipeline (Python logging) to monitor latency, throughput, and resource usage across distributed nodes with automated alerting.

Cybersecurity Research Intern Jun. 2024 – Aug. 2024
Zhejiang University *Hangzhou, China*

- Competitively selected as 1 of 6 students globally for a mobile application security research program focused on detecting insecure API usage in production Android apps.
- Automated a static analysis pipeline using Java, Soot, and FlowDroid to perform taint analysis and generate interprocedural control flow graphs (ICFGs), improving per-app audit time by 93% from 30 minutes to under 2 minutes
- Uncovered 50+ privacy policy violations by cross-referencing data-flow graphs against legal disclosures; presented findings at an international security research conference with 200+ attendees.

Projects

Racket Compiler | *Racket, C, x86-64 Assembly* Jan. – May 2025

- Built a multi-stage compiler from Racket to optimized x86-64 assembly, implementing tail-call optimization (TCO) to support deep recursion with constant stack usage.
- Designed a custom 64-bit calling convention and dynamic tagging system to enforce runtime type safety and arity validation with lower overhead than traditional boxing.
- Implemented a greedy register allocator to reduce stack spills, improving execution speed for compute-intensive recursive workloads.
- Wrote a C runtime using setjmp/longjmp to handle hardware-level exceptions and manage non-local control flow.

Video Performance Tracker | *Python, PostgreSQL, Docker, AWS* Dec. 2025

- Built an ETL pipeline in Python with SQLAlchemy to ingest engagement metrics from YouTube and TikTok APIs, centralizing data into a PostgreSQL database on AWS RDS.
- Developed a data collection service using Playwright with session management and proxy rotation to reliably scrape metrics from both platforms.
- Containerized the full stack with Docker Compose and deployed to AWS EC2; used persistent volumes and environment-separated configs to achieve identical dev/prod behavior and sub-10-minute deployment from scratch.

Personal Website | *React, Node.js, GitHub Actions, JavaScript, HTML* Jan. – May 2025

- Designed and deployed a personal portfolio using React and Node.js with a GitHub Actions CI/CD pipeline for automated builds and deployment to GitHub Pages.

Technical Skills

Languages: Python, Java, C, Rust, OCaml, JavaScript, TypeScript, SQL, x86-64 Assembly
Frameworks & Libraries: React, Node.js, AsyncIO, Pandas, Playwright, SQLAlchemy, FlowDroid
Infrastructure & Tools: Docker, AWS (EC2, RDS, S3), GitHub Actions, Linux, Git, PostgreSQL
Concepts: Compilers, Static/Taint Analysis, QUIC/UDP, Congestion Control, Parallel Computing, Register Allocation